



Exercise-1

✎ Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A) : Arithmetic Progression

- A-1.** In an A.P. the third term is four times the first term, and the sixth term is 17 ; find the series.
- A-2.** Find the sum of first 35 terms of the series whose p^{th} term is $\frac{p}{7} + 2$.
- A-3.** Find the number of integers between 100 & 1000 that are divisible by 7
- A-4.** Find the sum of all those integers between 100 and 800 each of which on division by 16 leaves the remainder 7.
- A-5. ✎** The sum of first p -terms of an A.P. is q and the sum of first q terms is p , find the sum of first $(p + q)$ terms.
- A-6.** The sum of three consecutive numbers in A.P. is 27, and their product is 504, find them.
- A-7. ✎** The fourth power of the common difference of an arithmetic progression with integer entries is added to the product of any four consecutive terms of it. Prove that the resulting sum is the square of an integer.
- A-8.** If a, b, c are in A.P., then show that:
 (i) $a^2(b + c), b^2(c + a), c^2(a + b)$ are also in A.P.
 (ii) $b + c - a, c + a - b, a + b - c$ are in A.P.
- A-9.** There are n A.M's between 3 and 54, such that the 8th mean: $(n - 2)^{\text{th}}$ mean :: 3: 5. The value of n is.

Section (B) : Geometric Progression

- B-1.** The third term of a G.P. is the square of the first term. If the second term is 8, find its sixth term.
- B-2. ✎** The continued product of three numbers in G.P. is 216, and the sum of the products of them in pairs is 156; find the numbers
- B-3. ✎** The sum of infinite number of terms of a G.P. is 4 and the sum of their cubes is 192. Find the series.
- B-4.** The sum of three numbers which are consecutive terms of an A.P. is 21. If the second number is reduced by 1 & the third is increased by 1, we obtain three consecutive terms of a G.P., find the numbers.
- B-5.** If the $p^{\text{th}}, q^{\text{th}}$ & r^{th} terms of an AP are in GP. Find the common ratio of the GP.
- B-6.** If a, b, c, d are in G.P., prove that :
 (i) $(a^2 - b^2), (b^2 - c^2), (c^2 - d^2)$ are in G.P.
 (ii) $\frac{1}{a^2 + b^2}, \frac{1}{b^2 + c^2}, \frac{1}{c^2 + d^2}$ are in G.P.
- B-7.** Let five geometric means are inserted between $\frac{32}{3}$ and $\frac{243}{2}$ then find sum of all the geometric means.





Section (C) : Harmonic and Arithmetic Geometric Progression

- C-1.** Find the 4th term of an H.P. whose 7th term is $\frac{1}{20}$ and 13th term is $\frac{1}{38}$.
- C-2.** Insert three harmonic means between 1 and 7.
- C-3.** If $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$ and p, q, r are in A.P. then prove that x, y, z are in H.P.
- C-4.** If a^2, b^2, c^2 are in A.P. show that $b+c, c+a, a+b$ are in H.P.
- C-5.** If b is the harmonic mean between a and c, then prove that $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$.
- C-6.** Sum the following series
- (i) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots$ to n terms.
- (ii) $1 + \frac{3}{4} + \frac{7}{16} + \frac{15}{64} + \frac{31}{256} + \dots$ to infinity.
- C-7.** Find the sum of n terms of the series the rth term of which is $(2r+1)2^r$.

Section (D) : Relation between A.M., G.M., H.M

- D-1.** Using the relation $A.M. \geq G.M.$ prove that
- (i) $(x^2y + y^2z + z^2x)(xy^2 + yz^2 + zx^2) \geq 9x^2y^2z^2$. (x, y, z are positive real number)
- (ii) $(a+b) \cdot (b+c) \cdot (c+a) > abc$; if a, b, c are positive real numbers
- D-2.** If $x > 0$, then find greatest value of the expression $\frac{x^{100}}{1+x+x^2+x^3+\dots+x^{200}}$.
- D-3.** The H.M. between two numbers is $\frac{16}{5}$, their A.M. is A and G.M. is G. If $2A + G^2 = 26$, then find the numbers.
- D-4.** If a, b, c are positive real numbers and sides of the triangle, then prove that $(a+b+c)^3 \geq 27(a+b-c)(c+a-b)(b+c-a)$
- D-5.** If $a_i > 0$ for all $i = 1, 2, 3, \dots, n$ then prove that $(1+a_1+a_1^2)(1+a_2+a_2^2) \dots (1+a_n+a_n^2) \geq 3^n(a_1 a_2 a_3 \dots a_n)$

Section (E) : Summation of series

- E-1.** Find the sum to n-terms of the sequence.
- (i) $1 + 5 + 13 + 29 + 61 + \dots$ up to n terms
- (ii) $3 + 33 + 333 + 3333 + \dots$ up to n terms
- E-2.** $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$ to n terms.



- E-3.** (i) If $t_n = 3^n - 2^n$ then find $\sum_{n=1}^k t_n$.
- (ii) If $t_n = n(n+2)$ then find $\sum_{n=1}^k t_n$.
- (iii) Find the sum to n terms of the series $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots$
- (iv) $10^2 + 13^2 + 16^2 + \dots$ upto 10 terms
- (v) If $\sum_{r=1}^n I(r) = n(2n^2 + 9n + 13)$, then find the $\sum_{r=1}^n \sqrt{I(r)}$

E-4. Find the sum to n -terms of the sequence.

- (i) $\frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} + \dots$
- (ii) $1 \cdot 3 \cdot 2^2 + 2 \cdot 4 \cdot 3^2 + 3 \cdot 5 \cdot 4^2 + \dots$

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Arithmetic Progression

- A-1.** The first term of an A.P. of consecutive integer is $p^2 + 1$. The sum of $(2p + 1)$ terms of this series can be expressed as
 (A) $(p + 1)^2$ (B) $(2p + 1)(p + 1)^2$ (C) $(p + 1)^3$ (D) $p^3 + (p + 1)^3$
- A-2.** If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \dots + a_{23} + a_{24}$ is equal to
 (A) 909 (B) 75 (C) 750 (D) 900
- A-3.** If the sum of the first $2n$ terms of the A.P. 2, 5, 8, ..., is equal to the sum of the first n terms of the A.P. 57, 59, 61, ..., then n equals
 (A) 10 (B) 12 (C) 11 (D) 13
- A-4.** The sum of integers from 1 to 100 that are divisible by 2 or 5 is
 (A) 2550 (B) 1050 (C) 3050 (D) none of these
- A-5.** Let 6 Arithmetic means $A_1, A_2, A_3, A_4, A_5, A_6$ are inserted between two consecutive natural number a and b ($a > b$). If $A_1^2 - A_2^2 + A_3^2 - A_4^2 + A_5^2 - A_6^2$ is equal to prime number then 'b' is equal to
 (A) 1 (B) 2 (C) 3 (D) 4

Section (B) : Geometric Progression

- B-1.** The third term of a G.P is 4. The product of the first five terms is
 (A) 4^3 (B) 4^5 (C) 4^4 (D) 4
- B-2.** If S is the sum to infinity of a G.P. whose first term is 'a', then the sum of the first n terms is
 (A) $S \left(1 - \frac{a}{S}\right)^n$ (B) $S \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$ (C) $a \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$ (D) $S \left[1 - \left(1 - \frac{S}{a}\right)^n\right]$
- B-3.** For a sequence $\{a_n\}$, $a_1 = 2$ and $\frac{a_{n+1}}{a_n} = \frac{1}{3}$. Then $\sum_{r=1}^{20} a_r$ is
 (A) $\frac{20}{2} [4 + 19 \times 3]$ (B) $3 \left(1 - \frac{1}{3^{20}}\right)$ (C) $2 (1 - 3^{20})$ (D) $\left(1 - \frac{1}{3^{20}}\right)$



- B-4.** α, β be the roots of the equation $x^2 - 3x + a = 0$ and γ, δ the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in this order) form an increasing G.P., then
 (A) $a = 3, b = 12$ (B) $a = 12, b = 3$ (C) $a = 2, b = 32$ (D) $a = 4, b = 16$
- B-5.** One side of an equilateral triangle is 24 cm. The mid-points of its sides are joined to form another triangle whose mid-points are in turn joined to form still another triangle. This process continues indefinitely. Then the sum of the perimeters of all the triangles is
 (A) 144 cm (B) 212 cm (C) 288 cm (D) 172 cm
- B-6.** Let 3 geometric means G_1, G_2, G_3 are inserted between two positive number a and b such that $\frac{G_3 - G_2}{G_2 - G_1} = 2$, then $\frac{b}{a}$ equal to
 (A) 2 (B) 4 (C) 8 (D) 16

Section (C) : Harmonic and Arithmetic Geometric Progression

- C-1.** If the 3rd, 6th and last term of a H.P. are $\frac{1}{3}, \frac{1}{5}, \frac{3}{203}$ then the number of terms is equal to
 (A) 100 (B) 102 (C) 99 (D) 101
- C-2.** If a, b, c are in H.P. then the value of $\frac{b+a}{b-a} + \frac{b+c}{b-c}$ is
 (A) 1 (B) 3 (C) 4 (D) 2
- C-3.** If the roots of the equation $x^3 - 11x^2 + 36x - 36 = 0$ are in H.P. then the middle root is
 (A) an even number (B) a perfect square of an integer
 (C) a prime number (D) a composite number
- C-4.** Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are:
 (A) not in A.P./G.P./H.P. (B) in A.P.
 (C) in G.P. (D) in H.P.
- C-5.** If $3 + \frac{1}{4}(3+d) + \frac{1}{4^2}(3+2d) + \dots + \text{upto } \infty = 8$, then the value of d is :
 (A) 9 (B) 5 (C) 1 (D) 4
- C-6.** Let 'n' Arithmetic Means and 'n' Harmonic Means are inserted between two positive number 'a' and 'b'. If sum of all Arithmetic Means is equal to sum of reciprocal all Harmonic means, then product of numbers is
 (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 3
- C-7.** Let $a_1, a_2, a_3, \dots$ be in A.P. and $h_1, h_2, h_3, \dots$ in H.P. If $a_1 = 2 = h_1$ and $a_{30} = 25 = h_{30}$ then $(a_7 h_{24} + a_{14} h_{17})$ equal to :
 (A) 50 (B) 100 (C) 200 (D) 400
- C-8.** **Statement 1** : 3, 6, 12 are in G.P., then 9, 12, 18 are in H.P.
Statement 2 : If three consecutive terms of a G.P. are positive and if middle term is added in these terms, then resultant will be in H.P.
 (A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
 (B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
 (C) STATEMENT-1 is true, STATEMENT-2 is false
 (D) STATEMENT-1 is false, STATEMENT-2 is true



- C-9.** $S = 3^{10} + 3^9 + \frac{3^9}{4} + \frac{3^7}{2} + \frac{5 \cdot 3^6}{16} + \frac{3^6}{16} + \frac{7 \cdot 3^4}{64} + \dots$ upto infinite terms, then $\left(\frac{25}{36}\right)S$ equals to
 (A) 6^9 (B) 3^{10} (C) 3^{11} (D) $2 \cdot 3^{10}$
- C-10.** The sum of infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + \dots \infty$
 (A) 21 (B) 22 (C) 23 (D) 24

Section (D) : Relation between A.M., G.M., H.M

- D-1.** If $x \in \mathbb{R}$, the numbers $5^{1+x} + 5^{1-x}$, $a/2$, $25^x + 25^{-x}$ form an A.P. then 'a' must lie in the interval:
 (A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$
- D-2.** If A, G & H are respectively the A.M., G.M. & H.M. of three positive numbers a, b, & c, then the equation whose roots are a, b, & c is given by :
 (A) $x^3 - 3Ax^2 + 3G^3x - G^3 = 0$ (B) $x^3 - 3Ax^2 + 3(G^3/H)x - G^3 = 0$
 (C) $x^3 + 3Ax^2 + 3(G^3/H)x - G^3 = 0$ (D) $x^3 - 3Ax^2 - 3(G^3/H)x + G^3 = 0$
- D-3.** If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation:
 (A) $0 \leq M \leq 1$ (B) $1 \leq M \leq 2$ (C) $2 \leq M \leq 3$ (D) $3 \leq M \leq 4$
- D-4.** If $a + b + c = 3$ and $a > 0$, $b > 0$, $c > 0$, the greatest value of $a^2b^3c^2$.
 (A) $\frac{3^{10} \cdot 2^4}{7^7}$ (B) $\frac{3^9 \cdot 2^4}{7^7}$ (C) $\frac{3^9 \cdot 2^5}{7^7}$ (D) $\frac{3^{10} \cdot 2^5}{7^7}$
- D-5.** If P, Q be the A.M., G.M. respectively between any two rational numbers a and b, then $P - Q$ is equal to
 (A) $\frac{a-b}{a}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\left(\frac{\sqrt{a}-\sqrt{b}}{\sqrt{2}}\right)^2$

Section (E) : Summation of series

- E-1** If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then value of $1 + \frac{3}{2} + \frac{5}{3} + \dots + \frac{2n-1}{n}$ is
 (A) $2n - H_n$ (B) $2n + H_n$ (C) $H_n - 2n$ (D) $H_n + n$
- E-2.** **Statement 1** : The sum of the first 30 terms of the sequence 1, 2, 4, 7, 11, 16, 22, is 4520.
Statement 2 : If the successive differences of the terms of a sequence form an A.P., then general term of sequence is of the form $an^2 + bn + c$.
 (A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
 (B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
 (C) STATEMENT-1 is true, STATEMENT-2 is false
 (D) STATEMENT-1 is false, STATEMENT-2 is true



E-3. The value of $\sum_{r=1}^n \frac{1}{\sqrt{a+r} \cdot x + \sqrt{a+(r-1)} \cdot x}$ is

- (A) $\frac{n}{\sqrt{a} + \sqrt{a+nx}}$ (B) $\frac{n}{\sqrt{a} - \sqrt{a+nx}}$ (C) $\frac{\sqrt{a+nx} - \sqrt{a}}{2x}$ (D) $\frac{\sqrt{a} + \sqrt{a+nx}}{x}$

E-4. The value of $(1.1^2 + 3.2^2 + 5.3^2 + \dots + \text{upto } 10 \text{ terms})$ is equal to :

- (A) 6050 (B) 5965 (C) 5665 (D) 5385

PART - III : MATCH THE COLUMN

- | 1. | Column – I | Column – II |
|-----|---|-------------------|
| (A) | The coefficient of x^{49} in the product $(x-1)(x-3)(x-5)(x-7) \dots (x-99)$ | (p) -2500 |
| (B) | Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is | (q) 9 |
| (C) | The sum $\sum_{r=2}^{\infty} \frac{1}{r^2 - 1}$ is equal to: | (r) $\frac{3}{4}$ |
| (D) | The length, breadth, height of a rectangular box are in G.P. (length > breadth > height) The volume is 27, the total surface area is 78. Then the length is | (s) 6 |
-
- | 2. | Column – I | Column – II |
|-----|---|-------------|
| (A) | The value of xyz is $\frac{15}{2}$ or $\frac{18}{5}$ according as the series a, x, y, z, b are in an A.P. or H.P. then ' $a + b$ ' equals where a, b are positive integers. | (p) 2 |
| (B) | The value of $2^{\frac{1}{4}} 4^{\frac{1}{8}} 8^{\frac{1}{16}} \dots \infty$ is equal to | (q) 1 |
| (C) | If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$, where $k \in \mathbb{N}$, then k is equal to | (r) 3 |
| (D) | There are m A.M. between 1 and 31. If the ratio of the 7^{th} and $(m-1)^{\text{th}}$ means is $5 : 9$, then $\frac{m}{7}$ is equal to | (s) 4 |

Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. Given the sequence of numbers $x_1, x_2, x_3, \dots, x_{2013}$ which satisfy $\frac{x_1}{x_1+1} = \frac{x_2}{x_2+3} = \frac{x_3}{x_3+5} = \dots = \frac{x_{2013}}{x_{2013}+4025}$, nature of the sequence is
- (A) A.P. (B) G.P. (C) H.P. (D) A.G.P.
2. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. if $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is
- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$



3. If 1, 2, 3 ... are first terms; 1, 3, 5 ... are common differences and $S_1, S_2, S_3, \dots$ are sums of n terms of given p AP's; then $S_1 + S_2 + S_3 + \dots + S_p$ is equal to
 (A) $\frac{np(np+1)}{2}$ (B) $\frac{n(np+1)}{2}$ (C) $\frac{np(p+1)}{2}$ (D) $\frac{np(np-1)}{2}$
4. If the sum of n terms of a G.P. (with common ratio r) beginning with the p^{th} term is k times the sum of an equal number of terms of the same series beginning with the q^{th} term, then the value of k is:
 (A) $r^{p/q}$ (B) $r^{q/p}$ (C) r^{p-q} (D) r^{p+q}
5. Consider the sequence 2, 3, 5, 6, 7, 8, 10, 11, of all positive integer, then 2011th term of this sequence is
 (A) 2056 (B) 2011 (C) 2013 (D) 2060
6. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in AP and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in :
 (A) HP (B) Arithmetico-Geometric Progression
 (C) AP (D) GP
7. If $a_1, a_2, \dots$ are in H.P. and $f(k) = \sum_{r=1}^k (a_r - a_k)$, then $\frac{a_1}{f(1)}, \frac{a_2}{f(2)}, \dots, \frac{a_n}{f(n)}$ are in
 (A) A.P. (B) G.P. (C) H.P. (D) None of these
8. If $a_1, a_2, a_3, \dots, a_n$ are positive real numbers whose product is a fixed number c , then the minimum value of $a_1 + a_2 + a_3 + \dots + a_{n-1} + 2a_n$ is
 (A) $n(2c)^{1/n}$ (B) $(n+1)c^{1/n}$ (C) $2nc^{1/n}$ (D) $(n+1)(2c)^{1/n}$
9. The sum of the first n -terms of the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$ is $\frac{n(n+1)^2}{2}$, when n is even. When n is odd, the sum is
 (A) $\frac{n(n+1)^2}{4}$ (B) $\frac{n^2(n+2)}{4}$ (C) $\frac{n^2(n+1)}{2}$ (D) $\frac{n(n+2)^2}{4}$
10. Let T_r and S_r be the r^{th} term and sum up to r^{th} term of a series respectively. If for an odd number n , $S_n = n$ and $T_n = \frac{T_{n-1}}{n^2}$ then T_m (m being even) is
 (A) $\frac{2}{1+m^2}$ (B) $\frac{2m^2}{1+m^2}$ (C) $\frac{(m+1)^2}{2+(m+1)^2}$ (D) $\frac{2(m+1)^2}{1+(m+1)^2}$
11. If $1^2 + 2^2 + 3^2 + \dots + 2003^2 = (2003)(4007)(334)$ and
 (1) $(2003) + (2)(2002) + (3)(2001) + \dots + (2003)(1) = (2003)(334)(x)$, then x equals
 (A) 2005 (B) 2004 (C) 2003 (D) 2001
12. If $\sum_{r=1}^n t_r = \frac{n(n+1)(n+2)(n+3)}{8}$, then $\sum_{r=1}^n \frac{1}{t_r}$ equals
 (A) $-\left(\frac{1}{(n+1)(n+2)} - \frac{1}{2}\right)$ (B) $\left(\frac{1}{(n+1)(n+2)} - \frac{1}{2}\right)$
 (C) $\left(\frac{1}{(n+1)(n+2)} + \frac{1}{2}\right)$ (D) $\left(\frac{1}{(n-1)(n-2)} + \frac{1}{2}\right)$



13. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto $\infty = \frac{\pi^2}{6}$, then $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots =$
 (A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) $\pi^2/4$
14. If $S_n = \sum_{k=1}^{8n} (-1)^{\frac{k(k+1)(k+2)}{6}} (k)^2 + \sum_{k=1}^{8n} (-1)^{\frac{(k+2)(k+3)}{2}} (k^2) - 4 \sum_{k=1}^{8n} (8k-2)^2$ then the value of $-S_4$ is equal to :
 (A) 256 (B) 512 (C) 1024 (D) 2048

PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

1. A man arranges to pay off a debt of Rs. 3600 by 40 annual installments which form an arithmetic series. When 30 of the installments are paid he dies leaving a third of the debt unpaid. Find the value of the first installment.
2. In a circle of radius R a square is inscribed, then a circle is inscribed in the square, a new square in the circle and so on for n times. Find the ratio of the limit of the sum of areas of all the circles to the limit of the sum of areas of all the squares as $n \rightarrow \infty$
3. If the common difference of the A.P. in which $T_7 = 9$ and $T_1 T_2 T_7$ is least, is 'd' then d is—
4. The number of terms in an A.P. is even ; the sum of the odd terms is 24, sum of the even terms is 30, and the last term exceeds the first by $10\frac{1}{2}$; find 13th term of A.P.
5. Let AP(a, d) terms of an infinite arithmetic progression with first term 'a' and common difference 'd' ($d > 0$). If $AP(1, 3) \cap AP(3, 5) \cap AP(5, 7) = AP(m, n)$ then $|n - m|$ equals to
6. If $x > 0$, and $\log_2 x + \log_2 (\sqrt{x}) + \log_2 (\sqrt[4]{x}) + \log_2 (\sqrt[8]{x}) + \log_2 (\sqrt[16]{x}) + \dots = 4$, then value of $\frac{\left(\frac{x^2+3x+2}{x+2}\right) + 3x - \frac{x(x^3+1)}{(x+1)(x^2-x+1)} \log_2 8}{(x-1)(\log_2 3)(\log_3 4)(\log_4 5)(\log_5 2)}$ is
7. Given that α, γ are roots of the equation $Ax^2 - 4x + 1 = 0$ and β, δ the roots of the equation $Bx^2 - 6x + 1 = 0$, then find value of $(A + B)$, such that α, β, γ & δ are in H.P.
8. Find sum of first 6 term of the infinitely decreasing G.P. whose third term, three times the product of the first and fourth term and second term form an A.P. in the indicated order, with common difference equal to $1/8$.
9. If a, b, c are in GP, $a - b, c - a, b - c$ are in HP, then the value of $a + 4b + c$ is
10. $a, a_1, a_2, a_3, \dots, a_{2n}, b$ are in A.P. and $a, g_1, g_2, g_3, \dots, g_{2n}, b$ are in G.P. and h is the harmonic mean of a and b, if $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n + a_{n+1}}{g_n g_{n+1}} +$ is equal $\frac{Kn}{20h}$ to , then find value of K.
11. If the arithmetic mean of two numbers a & b ($0 < a < b$) is 6 and their geometric mean G and harmonic mean H satisfy the relation $G^2 + 3H = 48$. Then find the value of $(2a - b)$
12. If $S = \frac{5}{13} + \frac{55}{(13)^2} + \frac{555}{(13)^3} + \dots$ up to ∞ , then find the value of S.



13. ✖ If $\frac{5}{k} = 1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \dots \infty$, then find the value of k
14. If $x_i > 0$, $i = 1, 2, \dots, 50$ and $x_1 + x_2 + \dots + x_{50} = 50$, then find the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$.
15. If a_1, a_2, a_3, a_4 are positive real numbers such that $a_1 + a_2 + a_3 + a_4 = 15$ then find maximum value of $(a_1 + a_2)(a_3 + a_4)$.
16. If S_1, S_2, S_3 are the sums of first n natural numbers, their squares, their cubes respectively, then $\frac{S_3(1+8S_1)}{S_2^2}$ equal to
17. ✖ If $S = \frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \infty$, then find the value of S.

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. ✖ The interior angles of a polygon are in A.P. If the smallest angle is 120° & the common difference is 5° , then the number of sides in the polygon is :
 (A) 7 (B) 9 (C) 16 (D) 5
2. If $1, \log_y x, \log_z y, -15 \log_x z$ are in A.P., then
 (A) $z^3 = x$ (B) $x = y^{-1}$ (C) $z^{-3} = y$ (D) $x = y^{-1} = z^3$
3. If $a_1, a_2, \dots, a_n$ are distinct terms of an A.P., then
 (A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$
 (C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$
4. First three terms of the sequence $1/16, a, b, 1/6$ are in geometric series and last three terms are in harmonic series if
 (A) $a = \frac{1}{9}, b = \frac{1}{12}$ (B) $a = \frac{1}{12}, b = \frac{1}{9}$ (C) $a = 1, b = -\frac{1}{4}$ (D) $a = -\frac{1}{4}, b = 1$
5. ✖ Which of the following numbers is/are composite
 (A) 1111.....1 (91 digits) (B) 1111.....1 (81 digits)
 (C) 1111.....1 (75 digits) (D) 1111.....1 (105 digits)
6. Three numbers a, b, c between 2 and 18 are such that
 (i) their sum is 25
 (ii) the numbers 2, a, b, are in A.P.
 (iii) the number b, c, 18 are in G.P.
 then which of the following options are correct.
 (A) $a = 5$ (B) $b = 8$ (C) $b + c = 20$ (D) $a + b + c = 25$
7. Consider an infinite geometric series with first term 'a' and common ratio r. If the sum is 4 and the second term is $3/4$, then:
 (A) $a = \frac{7}{4}, r = \frac{3}{7}$ (B) $a = 2, r = \frac{3}{8}$ (C) $a = \frac{3}{2}, r = \frac{1}{2}$ (D) $a = 3, r = \frac{1}{4}$



8. For the series $2 + \left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) + \left((2\sqrt{2} - 1) + \frac{1}{2}\right) + \left((3\sqrt{2} - 2) + \frac{1}{2\sqrt{2}}\right) + \dots$
- (A) $S_n = \sqrt{2}(\sqrt{2} + n - 1) - n + \left(\frac{(2^{n/2} - 1)}{(\sqrt{2} - 1) 2^{\frac{n-1}{2}}}\right)$ (B) $T_n = \sqrt{2}(\sqrt{2} + n - 1) - n + \left(\frac{1}{2}\right)^{\frac{n-1}{2}}$
- (C) $S_n = \frac{n}{2} (3 + (n-1)\sqrt{2} - n) + \left(\frac{(2^{n/2} - 1)}{(\sqrt{2} - 1) 2^{\frac{n-1}{2}}}\right)$ (D) $S_n = \frac{n}{2} (3 + (n-1)\sqrt{2} - n) +$
9. If $a_k a_{k-1} + a_{k-1} a_{k-2} = 2a_k a_{k-2}$, $k \geq 3$ and $a_1 = 1$, here $S_p = \sum_{k=1}^p \frac{1}{a_k}$ and given that $\frac{S_{2p}}{S_p}$ does not depend on p then $\frac{1}{a_{2016}}$ may be
- (A) 4031 (B) 1 (C) 2016 (D) 2017/2
10. If $\frac{a_{k+1}}{a_k}$ is constant for every $k \geq 1$. If $n > m \Rightarrow a_n > a_m$ and $a_1 + a_n = 66$, $a_2 a_{n-1} = 128$ and $\sum_{i=1}^n a_i = 126$ then
- (A) $n = 6$ (B) $n = 5$ (C) $\frac{a_{k+1}}{a_k} = 2$ (D) $\frac{a_{k+1}}{a_k} = 4$
11. The sides of a right triangle form a G.P. The tangent of the smallest angle is
- (A) $\sqrt{\frac{\sqrt{5}+1}{2}}$ (B) $\sqrt{\frac{\sqrt{5}-1}{2}}$ (C) $\sqrt{\frac{2}{\sqrt{5}+1}}$ (D) $\sqrt{\frac{2}{\sqrt{5}-1}}$
12. If b_1, b_2, b_3 ($b_i > 0$) are three successive terms of a G.P. with common ratio r , the value of r for which the inequality $b_3 > 4b_2 - 3b_1$ holds is given by
- (A) $r > 3$ (B) $0 < r < 1$ (C) $r = 3.5$ (D) $r = 5.2$
13. If a satisfies the equation $a^{2017} - 2a + 1 = 0$ and $S = 1 + a + a^2 + \dots + a^{2016}$. then possible value(s) of S is/are
- (A) 2016 (B) 2018 (C) 2017 (D) 2
14. Let two term of H.P. are 12 and 15. All terms of this H.P. Consist of natural number. If among the largest possible such H.P., the three of the terms are a, b, c which are increasing G.P. then
- (A) $a = 15$ (B) $b = 30$ (C) $c = 60$ (D) $a + b + c = 2$
15. Let a, x, b be in A.P.; a, y, b be in G.P and a, z, b be in H.P. If $x = y + 2$ and $a = 5z$, then
- (A) $y^2 = xz$ (B) $x > y > z$ (C) $a = 9, b = 1$ (D) $a = 1/4, b = 9/4$
16. Let $x, a_1, a_2, \dots, a_n, y, a_{n+1}, a_{n+2}, \dots, a_{2n}, z$ are in A.P.
 $p, g_1, g_2, \dots, g_n, q, g_{n+1}, g_{n+2}, \dots, g_{2n}, r$ are in G.P.
 $k, h_1, h_2, \dots, h_n, l, h_{n+1}, h_{n+2}, \dots, h_{2n}, m$ are in H.P.
- (A) If $x = p = k = 20$ and $y = q = \ell = 30$ then $z < r < m$
 (B) If $y = q = \ell = 20$ and $z = r = m = 30$ then $x < p < k$
 (C) If $x = p = k = 20$ and $z = r = m = 30$ then $y > q > \ell$
 (D) If $x = p = k = 20$ and $z = r = m = 30$ then $y < q < \ell$



17. Which of the following is/are TRUE
- (A) Equal numbers are always in A.P. , G.P. and H.P.
- (B) If a, b, c be in H.P., then $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ will be in AP
- (C) If G_1 and G_2 are two geometric means and A is the arithmetic mean inserted between two positive numbers, then the value of $\frac{G_1^2}{G_2} + \frac{G_2^2}{G_1}$ is $2A$.
- (D) Let general term of a G.P. (with positive terms) with common ratio r be T_{k+1} and general term of another G.P. (with positive terms) with common ratio r be T'_{k+1} , then the series whose general term $T''_{k+1} = T_{k+1} + T'_{k+1}$ is also a G.P. with common ratio r .
18. If the arithmetic mean of two positive numbers a & b ($a > b$) is twice their geometric mean, then $a : b$ is:
- (A) $2 + \sqrt{3} : 2 - \sqrt{3}$ (B) $7 + 4\sqrt{3} : 1$ (C) $1 : 7 - 4\sqrt{3}$ (D) $2 : \sqrt{3}$
19. If $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then
- (A) $a + c = b + d$ (B) $e = 0$
 (C) $a, b - 2/3, c - 1$ are in A.P. (D) c/a is an integer
20. The roots of the equation $x^4 - 8x^3 + ax^2 - bx + 16 = 0$, are positive, if
- (A) $a = 24$ (B) $a = 12$ (C) $b = 8$ (D) $b = 32$
21. Let $a_1, a_2, a_3, \dots, a_n$ is the sequence whose sum of first 'n' terms is represented by $S_n = an^3 + bn^2 + cn, n \in \mathbb{N}$. If $a = \frac{a_1 + a_3 - xa_2}{y}$ then
- (A) H.C.F of (x, y) is 2 (B) H.C.F. of (x, y) is 3
 (C) L.C.M of (x, y) is 6 (D) $x + y = 8$

PART - IV : COMPREHENSION

Comprehension # 1 (Q.1 & 2)

We know that $1 + 2 + 3 + \dots = \frac{n(n+1)}{2} = f(n)$,

$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = g(n)$,

$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = h(n)$

1. Even natural number which divides $g(n) - f(n)$, for every $n \geq 2$, is
- (A) 2 (B) 4 (C) 6 (D) none of these
2. $f(n) + 3g(n) + h(n)$ is divisible by $1 + 2 + 3 + \dots + n$
- (A) only if $n = 1$ (B) only if n is odd (C) only if n is even (D) for all $n \in \mathbb{N}$

Comprehension # 2 (Q.3 & 4)

In a sequence of $(4n + 1)$ terms the first $(2n + 1)$ terms are in AP whose common difference is 2, and the last $(2n + 1)$ terms are in GP whose common ratio 0.5. If the middle terms of the AP and GP are equal, then

3. Middle term of the sequence is
- (A) $\frac{n \cdot 2^{n+1}}{2^n - 1}$ (B) $\frac{n \cdot 2^{n+1}}{2^{2n} - 1}$ (C) $n \cdot 2^n$ (D) None of these



4. First term of the sequence is

(A) $\frac{4n+2n \cdot 2^n}{2^n-1}$ (B) $\frac{4n-2n \cdot 2^n}{2^n-1}$ (C) $\frac{2n-n \cdot 2^n}{2^n-1}$ (D) $\frac{2n+n \cdot 2^n}{2^n-1}$

Comprehension # 3 (Q.5 to 7)

Let $\Delta^1 T_n = T_{n+1} - T_n$, $\Delta^2 T_n = \Delta^1 T_{n+1} - \Delta^1 T_n$, $\Delta^3 T_n = \Delta^2 T_{n+1} - \Delta^2 T_n$, , and so on, where $T_1, T_2, T_3, \dots, T_{n-1}, T_n, T_{n+1}, \dots$ are the terms of infinite G.P. whose first term is a natural number and common ratio is equal to 'r'.

5. If $\Delta^2 T_1 = 36$, then sum of all possible integral values of r is equal to :
 (A) 8 (B) 4 (C) 5 (D) -2
6. Let $\sum_{n=1}^{\infty} T_n = \frac{7}{3}$ and $r = \frac{p}{7}$ then sum of squares of all possible value of p is equal to :
 (A) 42 (B) 46 (C) 45 (D) 30
7. If $\Delta^7 T_n = \Delta^3 T_n$, then 'r' equal to
 (A) 2 (B) 4 (C) 7 (D) -2

Exercise-3

Marked questions are recommended for Revision.

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

1. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is
 [IIT-JEE - 2009, Paper-2, (3, -1), 80]
 (A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$
2. Let S_k , $k = 1, 2, \dots, 100$, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} + \sum_{k=1}^{100} (k^2 - 3k + 1) S_k$ is
 [IIT-JEE - 2010, Paper-1, (3, 0), 84]
3. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$. If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to
 [IIT-JEE - 2010, Paper-2, (3, 0), 79]
4. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$.
 For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is
 [IIT-JEE 2011, Paper-1, (4, 0), 80]
5. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} where $a > 0$ is
 [IIT-JEE 2011, Paper-1, (4, 0), 80]



6. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is
[IIT-JEE 2012, Paper-2, (3, -1), 66]
 (A) 22 (B) 23 (C) 24 (D) 25
7. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) **[JEE (Advanced) 2013, Paper-1, (4, -1)/60]**
 (A) 1056 (B) 1088 (C) 1120 (D) 1332
8. A pack contains n card numbered from 1 to n . Two consecutive numbered card are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k , then $k - 20 =$ **[JEE (Advanced) 2013, Paper-1, (4, -1)/60]**
9. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is **[JEE (Advanced) 2014, Paper-1, (3, 0)/60]**
10. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE (Advanced) 2015, P-2 (4, 0) / 80]**
11. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is **[JEE (Advanced) 2016, Paper-1, (3, -1)/62]**
 (A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{27}$ (D) $\frac{1}{25}$
12. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$, then **[JEE (Advanced) 2016, Paper-2, (3, -1)/62]**
 (A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
 (C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$
13. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side ? **[JEE(Advanced) 2017, Paper-1, (3, 0)/61]**
14. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, Then, the number of elements in the set $X \cup Y$ is _____. **[JEE(Advanced) 2018, Paper-1, (3, 0)/60]**
15. Let AP (a, d) denote the set of all the term of an infinite arithmetic progression with first term a and common difference $d > 0$. If $AP(1, 3) \cap AP(2, 5) \cap AP(3, 7) = AP(a, d)$ then $a + d$ equals..... **[JEE(Advanced) 2019, Paper-1, (4, -1)/62]**

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in an AP with common difference -2 , then the time taken by him to count all notes is **[AIEEE 2010 (8, -2), 144]**
 (1) 34 minutes (2) 125 minutes (3) 135 minutes (4) 24 minutes



2. A man saves Rs. 200 in each of the first three months of his service. In each of the subsequent months his saving increases by Rs. 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs. 11040 after : **[AIEEE 2011, I, (4, -1), 120]**
 (1) 18 months (2) 19 months (3) 20 months (4) 21 months
3. Let a_n be the n^{th} term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is : **[AIEEE 2011, II, (4, -1), 120]**
 (1) $\alpha - \beta$ (2) $\frac{\alpha - \beta}{100}$ (3) $\beta - \alpha$ (4) $\frac{\alpha - \beta}{200}$
4. The sum of first 20 terms of the sequence 0.7, 0.77, 0.777,....., is **[AIEEE - 2013, (4, -1), 360]**
 (1) $\frac{7}{81} (179 - 10^{-20})$ (2) $\frac{7}{9} (99 - 10^{-20})$ (3) $\frac{7}{81} (179 + 10^{-20})$ (4) $\frac{7}{9} (99 + 10^{-20})$
5. If $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10 (11)^9 = k(10)^9$, then k is equal to **[JEE(Main) 2014, (4, -1), 120]**
 (1) 100 (2) 110 (3) $\frac{121}{10}$ (4) $\frac{441}{100}$
6. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is **[JEE(Main) 2014, (4, -1), 120]**
 (1) $2 - \sqrt{3}$ (2) $2 + \sqrt{3}$ (3) $\sqrt{2} + \sqrt{3}$ (4) $3 + \sqrt{2}$
7. If m is the A. M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals : **[JEE(Main) 2015, (4, -1), 120]**
 (1) $4 l^2 m n$ (2) $4 l m^2 n$ (3) $4 l m n^2$ (4) $4 l^2 m^2 n^2$
8. The sum of first 9 terms of the series $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ is : **[JEE(Main) 2015, (4, -1), 120]**
 (1) 71 (2) 96 (3) 142 (4) 192
9. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is : **[JEE(Main) 2016, (4, -1), 120]**
 (1) $\frac{4}{3}$ (2) 1 (3) $\frac{7}{4}$ (4) $\frac{8}{5}$
10. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$ is $\frac{16}{5}m$, then m is equal to : **[JEE(Main) 2016, (4, -1), 120]**
 (1) 101 (2) 100 (3) 99 (4) 102
11. For any three positive real numbers a, b and c, $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$, Then **[JEE(Main) 2017, (4, -1), 120]**
 (1) b, c and a are in G.P. (2) b, c and a are in A.P.
 (3) a, b and c are in A.P. (4) a, b and c are in G.P.
12. Let $a, b, c \in \mathbb{R}$. If $f(x) = ax^2 + bx + c$ is such that $a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy, \forall x, y \in \mathbb{R}$, then $\sum_{n=1}^{10} f(n)$ is equal to **[JEE(Main) 2017, (4, -1), 120]**
 (1) 330 (2) 165 (3) 190 (4) 225
13. If, for a positive integer n, the quadratic equation, $x(x + 1) + (x + 1)(x + 2) + \dots + (x + n - 1)(x + n) = 10n$ has two consecutive integral solutions, then n is equal to **[JEE(Main) 2017, (4, -1), 120]**
 (1) 12 (2) 9 (3) 10 (4) 11



14. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$, then m is equal to : **[JEE(Main) 2018, (4, -1), 120]**
 (1) 34 (2) 33 (3) 66 (4) 68
15. Let A be the sum of the first 20 terms and B be sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to : **[JEE(Main) 2018, (4, -1), 120]**
 (1) 464 (2) 496 (3) 232 (4) 248
16. The sum of the following series $1 + 6 + \frac{9(1^2 + 2^2 + 3^2)}{7} + \frac{12(1^2 + 2^2 + 3^2 + 4^2)}{9} + \frac{15(1^2 + 2^2 + \dots + 5^2)}{11} + \dots$ up to 15 terms, is : **[JEE(Main) 2019, Online (09-01-19), P-2 (4, -1), 120]**
 (1) 7510 (2) 7830 (3) 7520 (4) 7820
17. If $5, 5r, 5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to : **[JEE(Main) 2019, Online (10-01-19), P-1 (4, -1), 120]**
 (1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $\frac{7}{4}$ (4) $\frac{5}{4}$
18. The sum of all natural number 'n' such that $100 < n < 200$ and $\text{H.C.F.}(91, n) > 1$ is : **[JEE(Main) 2019, Online (08-04-19), P-1 (4, -1), 120]**
 (1) 3303 (2) 3203 (3) 3221 (4) 3121
19. Let $a_1, a_2, a_3, \dots$ be an A.P. with $a_6 = 2$. Then the common difference of this A.P., which maximizes the product $a_1 a_4 a_5$ is : **[JEE(Main) 2019, Online (10-04-19), P-2 (4, -1), 120]**
 (1) $\frac{3}{2}$ (2) $\frac{8}{5}$ (3) $\frac{6}{5}$ (4) $\frac{2}{3}$
20. Let a, b and c be in G.P. with common ratio r , where $a \neq 0$ and $0 < r \leq \frac{1}{2}$. If $3a, 7b$ and $15c$ are the first three terms of an A.P., then the 4th term of this A.P. is : **[JEE(Main) 2019, Online (10-04-19), P-2 (4, -1), 120]**
 (1) $\frac{7}{3}a$ (2) a (3) $\frac{2}{3}a$ (4) $5a$
21. The greatest positive integer k , for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$, is **[JEE(Main) 2020, Online (07-01-20), P-1 (4, -1), 120]**
 (1) 32 (2) 63 (3) 65 (4) 60
22. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that for all $x \in \mathbb{R}$ $(2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is : **[JEE(Main) 2020, Online (08-01-20), P-1 (4, -1), 120]**
 (1) 0 (2) 4 (3) 4 (4) 3
23. Let a_n be the n^{th} term of a G.P. of positive terms. If $\sum_{n=1}^{100} a_{2n+1} = 200$ and $\sum_{n=1}^{100} a_{2n} = 100$, then $\sum_{n=1}^{200} a_n$ is equal to : **[JEE(Main) 2020, Online (09-01-20), P-2 (4, -1), 120]**
 (1) 175 (2) 150 (3) 300 (4) 225
24. If $x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n} \theta$ and $y = \sum_{n=0}^{\infty} \cos^{2n} \theta$, for $0 < \theta < \frac{\pi}{4}$, then:
 (1) $y(1-x) = 1$ (2) $y(1+x) = 1$ (3) $x(1+y) = 1$ (4) $x(1-y) = 1$



Answers

EXERCISE - 1

PART - I

Section (A) :

- A-1. 2, 5, 8, A-2. 160 A-3. 128 A-4. 19668 A-5. $-(p + q)$ A-6. 4, 9, 14
A-9. 16

Section (B) :

- B-1. 128 B-2. 2, 6, 18 or 18, 6, 2 B-3. 6, -3, $3/2$, B-4. 3, 7, 11 or 12, 7, 2
B-5. $\frac{q-r}{p-q}$ B-7. 211

Section (C) :

- C-1. $\frac{1}{11}$ C-2. $\frac{14}{11}, \frac{14}{8}, \frac{14}{5}$ C-6. (i) $4 - \frac{2+n}{2^{n-1}}$ (ii) $\frac{8}{3}$
C-7. $n \cdot 2^{n+2} - 2^{n+1} + 2$

Section (D) :

- D-2. $\frac{1}{201}$ D-3. 2, 8

Section (E) :

- E-1. (i) $2^{n+2} - 3n - 4$ (ii) $\frac{1}{27} (10^{n+1} - 9n - 10)$ E-2. $\frac{n \cdot 2^n - 2^n + 1}{2^n}$
E-3. (i) $\frac{1}{2} (3^{k+1} + 1) - 2^{k+1}$ (ii) $\frac{1}{6} k(k+1)(2k+7)$
(iii) $-\frac{n(n+1)}{2}$ if n is even, $\frac{n(n+1)}{2}$ if n is odd (iv) 6265 (v) $\sqrt{\frac{3}{2}}(n^2 + 3n)$
E-4. (i) $\frac{1}{12} - \frac{1}{4(2n+1)(2n+3)}$ (ii) $\frac{n}{10} (n+1)(n+2)(n+3)(2n+3)$

PART - II

Section (A) :

- A-1. (D) A-2. (D) A-3. (C) A-4. (C) A-5. (C)

Section (B) :

- B-1. (B) B-2. (B) B-3. (B) B-4. (C) B-5. (A) B-6. (D)

Section (C) :

- C-1. (A) C-2. (D) C-3. (C) C-4. (D) C-5. (A) C-6. (A) C-7. (B)
C-8. (A) C-9. (B) C-10. (C)

Section (D) :

- D-1. (D) D-2. (B) D-3. (A) D-4. (A) D-5. (D)

Section (E) :

- E-1. (A) E-2. (D) E-3. (A) E-4. (C)

PART - III

1. (A) $\rightarrow$ (p), (B) $\rightarrow$ (s), (C) $\rightarrow$ (r), (D) $\rightarrow$ (q)
2. (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)



EXERCISE - 2

PART - I

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (A) | 2. (D) | 3. (A) | 4. (C) | 5. (A) |
| 6. (A) | 7. (C) | 8. (A) | 9. (C) | 10. (D) |
| 11. (A) | 12. (A) | 13. (C) | 14. (C) | |

PART - II

- | | | | | |
|-------------------|-----------|-------------------|-----------|-----------|
| 1. 51.00 | 2. 00.78 | 3. 01.65 | 4. 19.50 | 5. 02.00 |
| 6. 01.66 or 01.67 | 7. 11.00 | 8. 01.96 or 01.97 | 9. 00.00 | 10. 40.00 |
| 11. 00.00 | 12. 01.80 | 13. 10.80 | 14. 50.00 | 15. 56.25 |
| 16. 09.00 | 17. 00.50 | | | |

PART - III

- | | | | | |
|-----------|------------|-----------|------------|-----------|
| 1. (B) | 2. (ABCD) | 3. (BD) | 4. (BD) | 5. (ABCD) |
| 6. (ABCD) | 7. (D) | 8. (BC) | 9. (AB) | 10. (AC) |
| 11. (BC) | 12. (ABCD) | 13. (CD) | 14. (ABC) | 15. (ABC) |
| 16. (ABC) | 17. (CD) | 18. (ABC) | 19. (ABCD) | 20. (AD) |
| 21. (ACD) | | | | |

PART - IV

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (A) | 2. (D) | 3. (A) | 4. (B) | 5. (A) |
| 6. (B) | 7. (A) | | | |

EXERCISE - 3

PART - I

- | | | | | |
|-----------|--|----------|---------|------------|
| 1. (C) | 2. 3 | 3. 0 | | |
| 4. | 3 or 9, both 3 and 9 (The common difference of the arithmetic progression can be either 0 or 6, and accordingly the second term can be either 3, or 9 ; thus the answers 3, or 9, or both 3 and 9 are acceptable.) | | | |
| 5. (8) | 6. (D) | 7. (A,D) | 8. 5 | 9. (4) |
| 10. 9 | 11. (C) | 12. (B) | 13. (6) | 14. (3748) |
| 15. (157) | | | | |

PART - II

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (1) | 2. (4) | 3. (2) | 4. (3) | 5. (1) |
| 6. (2) | 7. (2) | 8. (2) | 9. (1) | 10. (1) |
| 11. (2) | 12. (1) | 13. (4) | 14. (1) | 15. (4) |
| 16. (4) | 17. (3) | 18. (4) | 19. (2) | 20. (2) |
| 21. (2) | 22. (4) | 23. (2) | 24. (1) | |



High Level Problems (HLP)

SUBJECTIVE QUESTIONS

1. Prove that $\sqrt{2}, \sqrt{3}, \sqrt{5}$ cannot be terms of a single A.P.
2. If the sum of the first m terms of an A.P. is equal to the sum of either the next n terms or the next p terms, then prove that $(m+n) \left(\frac{1}{m} - \frac{1}{p} \right) = (m+p) \left(\frac{1}{m} - \frac{1}{n} \right)$.
3. If a and b are p^{th} and q^{th} terms of an AP, then find the sum of its $(p+q)$ terms
4. In an A.P. of which ' a ' is the 1st term, if the sum of the 1st ' p ' terms is equal to zero, show that the sum of the next ' q ' terms is $-\frac{a(p+q)q}{p-1}$.
5. If $\frac{a+be^y}{a-be^y} = \frac{b+ce^y}{b-ce^y} = \frac{c+de^y}{c-de^y}$, then show that a, b, c, d are in G.P.
6. The sum of the first ten terms of an AP is 155 & the sum of first two terms of a GP is 9. The first term of the AP is equal to the common ratio of the GP & the first term of the GP is equal to the common difference of the AP. Find the two progressions.
7. Find the sum in the n^{th} group of sequence,
(i) (1), (2, 3); (4, 5, 6, 7); (8, 9, ..., 15);
(ii) (1), (2, 3, 4), (5, 6, 7, 8, 9),
8. Let a, b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, show that $\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a+b)(a+2b)}{9ab}$.
9. If total number of runs scored in n matches is $\left(\frac{n+1}{4} \right) (2^{n+1} - n - 2)$ where $n > 1$ and the runs scored in the k^{th} match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$, find n
10. Let $a_1, a_2, \dots, a_n$ be positive real numbers in geometric progression. For each n , let A_n, G_n, H_n be respectively the arithmetic mean, geometric mean & harmonic mean of $a_1, a_2, \dots, a_n$. Prove that $G = \prod_{k=1}^n (A_k H_k)^{\frac{1}{2n}}$, Where G is geometric mean between $G_1, G_2, \dots, G_n$.
11. If a, b, c are in A.P., p, q, r are in H.P. and ap, bq, cr are in G.P., then find $\frac{p}{r} + \frac{r}{p}$.
12. If the sum of the roots of the quadratic equation, $ax^2 + bx + c = 0$ is equal to sum of the squares of their reciprocals, then prove that $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in H.P.
13. If a, b, c are in H.P.; b, c, d are in G.P.; and c, d, e are in A.P. such that $(ka-b)^2 e = ab^2$ then value of k .



14. The value of $x + y + z$ is 15 if a, x, y, z, b are in AP while the value of $(1/x) + (1/y) + (1/z)$ is $5/3$ if a, x, y, z, b are in HP. Find a and b .
15. If n is a root of the equation $x^2(1 - ac) - x(a^2 + c^2) - (1 + ac) = 0$ and if n HM's are inserted between a and c , show that the difference between the first and the last mean is equal to $ac(a - c)$.
16. If a, b, c are positive real numbers, then prove that
 (i) $b^2c^2 + c^2a^2 + a^2b^2 \geq abc(a + b + c)$.
 (ii) $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$
 (iii) $\frac{2}{a+b} + \frac{2}{b+c} + \frac{2}{c+a} \geq \frac{9}{a+b+c}$
17. Solve the equation $(2 + x_1 + x_2 + x_3 + x_4)^5 = 6250 x_1 x_2 x_3 x_4$ where $x_1, x_2, x_3, x_4 > 0$.
18. Let $a_1, a_2, \dots, a_n$ be real numbers such that

$$\sqrt{a_1} + \sqrt{a_2 - 1} + \sqrt{a_3 - 2} + \dots + \sqrt{a_n - (n-1)} = \frac{1}{2} (a_1 + a_2 + \dots + a_n) - \frac{n(n-3)}{4}$$

 then find the value of $\sum_{i=1}^{100} a_i$
19. If $a_i \in \mathbb{R}, i = 1, 2, 3, \dots, n$ and all a_i 's are distinct such that $\left(\sum_{i=1}^{n-1} a_i^2\right) + 6\left(\sum_{i=1}^{n-1} a_i - a_{i+1}\right) + 9\sum_{i=2}^n a_i^2 \leq 0$
 and $a_1 = 8$ then find the sum of first five terms.
20. Let $\{a_n\}$ and $\{b_n\}$ are two sequences given by $a_n = (x)^{1/2^n} + (y)^{1/2^n}$ and $b_n = (x)^{1/2^n} - (y)^{1/2^n}$ for all $n \in \mathbb{N}$.
 Then find $a_1 a_2 a_3 \dots a_n$.
21. Given that $a_1, a_2, a_3, \dots, a_n$ form an A.P. find then following sum $\sum_{i=1}^{10} \frac{a_i a_{i+1} a_{i+2}}{a_i + a_{i+2}}$
 Given that $a_1 = 1; a_2 = 2$
22. Find sum of the series $\frac{n}{1 \cdot 2 \cdot 3} + \frac{n-1}{2 \cdot 3 \cdot 4} + \frac{n-2}{3 \cdot 4 \cdot 5} + \dots$ up to n terms..
23. Find the value of $S_n = \sum_{r=1}^n \frac{3^n \cdot 5^n}{(5^n - 3^n)(5^{n+1} - 3^{n+1})}$ and hence S_∞ .
24. Circles are inscribed in the acute angle α so that every neighbouring circles touch each other. If the radius of the first circle is R , then find the sum of the radii of the first n circles in terms of R and α .
25. Let A, G, H be A.M., G.M. and H.M. of three positive real numbers a, b, c respectively such that $G^2 = AH$, then prove that a, b, c are terms of a GP.
26. If $S_n = \sum_{r=1}^n t_r = \frac{1}{6} n (2n^2 + 9n + 13)$, then $\sum_{r=1}^{\infty} \frac{1}{r \cdot \sqrt{t_r}}$ equals
27. In the quadratic equation $ax^2 + bx + c = 0, a \neq 0, \Delta = b^2 - 4ac$ and $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P. where α, β are the root of $ax^2 + bx + c = 0$, then prove that $c\Delta = 0$
28. If sum of first n terms of an A.P. (having positive terms) is given by $S_n = (1 + 2T_n)(1 - T_n)$ where T_n is the n^{th} term of series, then $T_2^2 = \frac{\sqrt{a} - \sqrt{b}}{4}, (a \in \mathbb{N}, b \in \mathbb{N})$, then find the value of $(a + b)$



Answers

3. $\frac{p+q}{2} \left[a+b+\frac{a-b}{p-q} \right]$
6. $(3 + 6 + 12 + \dots) ; (2/3 + 25/3 + 625/6 + \dots)$ G.P.
 $(2 + 5 + 8 + \dots) ; \left(\frac{25}{2} + \frac{79}{6} + \frac{83}{6} + \dots \right)$ A.P.
7. (i) $2^{n-2} (2^n + 2^{n-1} - 1)$ (ii) $(n-1)^3 + n^3$
9. 7
11. $\frac{a}{c} + \frac{c}{a}$
13. 2
14. $a = 1, b = 9$ OR $b = 1, a = 9$
18. 5050
19. $\frac{488}{81}$
20. $\frac{x-y}{b_n}$
21. $\frac{495}{2}$
22. $\frac{n(n+1)}{4(n+2)}$
23. $\frac{3}{4}$
24. $\frac{R \left(1 - \sin \frac{\alpha}{2} \right)}{2 \sin \frac{\alpha}{2}} \left[\left(\frac{1 + \sin \frac{\alpha}{2}}{1 - \sin \frac{\alpha}{2}} \right)^n - 1 \right]$
26. 1
28. 6